

International Institute of Information Technology, Hyderabad
(Deemed to be University)

CS6.201: Introduction to Software Systems; Spring 2026
IIIT Hyderabad Mid Term Exam

Max. Time: 1.5 Hrs

Max. Marks: 100

Q1. [2 × 7 = 14 marks] Give the git commands for the following:

- a) Initialize git in a new directory and list un-staged changes in working directory.
- b) Undo the changes in working directory of one specific file (blame.txt).
- c) Move all changes from working directory to staging area.
- d) Create a new branch with staged or unstaged changes.
- e) Move a branch to a specific commit, while removing changes from staged area but keeps changes in working directory
- f) Update the last local commit
- g) Undo a commit that has already been pushed to a shared remote repository

Q2. [6 + 5 = 11 marks] Write **bash** commands to do the following:

- a) Count the number of executable .py files in the current directory.
- b) Decide if a particular name is that of a directory or not.

Q3. [9 + 6 = 15 marks] **Python, Problem Solving.**

- a) Write a python function, RelPrime(i1, i2), that returns **True** if the integers i1 and i2 are relatively prime (no common factors), **False** otherwise.
- b) A sortedInt is a sorting of the digits of an integer in descending order (sortedInt(14524) is 54421). Using (a), write a python script that takes as input, an integer and prints if the integer and its sortedInt are relatively prime or not.

Q4. [8 + 7 = 15 marks] **Files and Text Processing.**

- a) Write a python program that reads in a text file and find the top three most frequent words in it.
- b) What are the time and space complexities of your solution? State your assumptions. How can the performance be improved?

Q5. [7 + 8 = 15 marks] **Python Internals.**

- a) State three differences between mutable and immutable data types in Python? Give examples of each type.
- b) Dictionaries are implemented as hash tables in Python. Suggest an alternative implementation along with a list of advantages and disadvantages of your approach.

Q6. [7 + 8 = 15 marks] **Python Lambda Functions.**

- a) What is a lambda function and when do you prefer it over a regular function?
- b) Write a lambda function that checks if an integer is a perfect square or not.

Q7. [7 + 8 = 15 marks] **Python List Comprehensions.**

- a) What is List Comprehension in Python and how is it different from a map function?
- b) Given a list of verbs, use list comprehension to create a new list of past-tense verbs (ending in 'ed') after converting them to present-tense (remove ed from the end).